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AIM: To evaluate the impact of varying k-mer sizes (k=21, 31, 41) on *de novo* bacterial genome assembly contiguity, metrics, and gene content completeness using 1 million paired-end short reads.

EXPERIMENTAL DESIGN

Sample / Dataset	**Biological Condition / Source**	**Read Type**	**Sequencing Platform**
SR1_1M.fastq	Bacterial Paired-End Subsample	150 bp Paired-End	Illumina NovaSeq 6000
SR2_1M.fastq	Bacterial Paired-End Subsample	150 bp Paired-End	Illumina NovaSeq 6000

WORKFLOW SUMMARY

Step 1: Environment Initialization: Deploy the required bioinformatics tool suite within an isolated Pixi environment.

Step 2: Read Quality Inspection: Evaluate raw base quality scores, sequence composition, and adapter content using FastQC.

Step 3: De Bruijn Graph Construction: Hash input sequences into k-mers and build assembly graphs across varying k-mer sizes (k=21, 31, 41) using Velveth.

Step 4: De Novo Assembly Execution: Traverse the constructed De Bruijn graphs and simplify paths to generate genomic contigs using Velvetg.

Step 5: Assembly Metric Benchmarking: Quantify contiguity, N50 metrics, and total assembly span across all k-mer sizes using QUAST.

Step 6: Gene Completeness Evaluation: Benchmarking single-copy ortholog completeness across assemblies using BUSCO.

DETAILED METHODOLOGY, COMMAND ANALYSIS, AND TROUBLESHOOTING

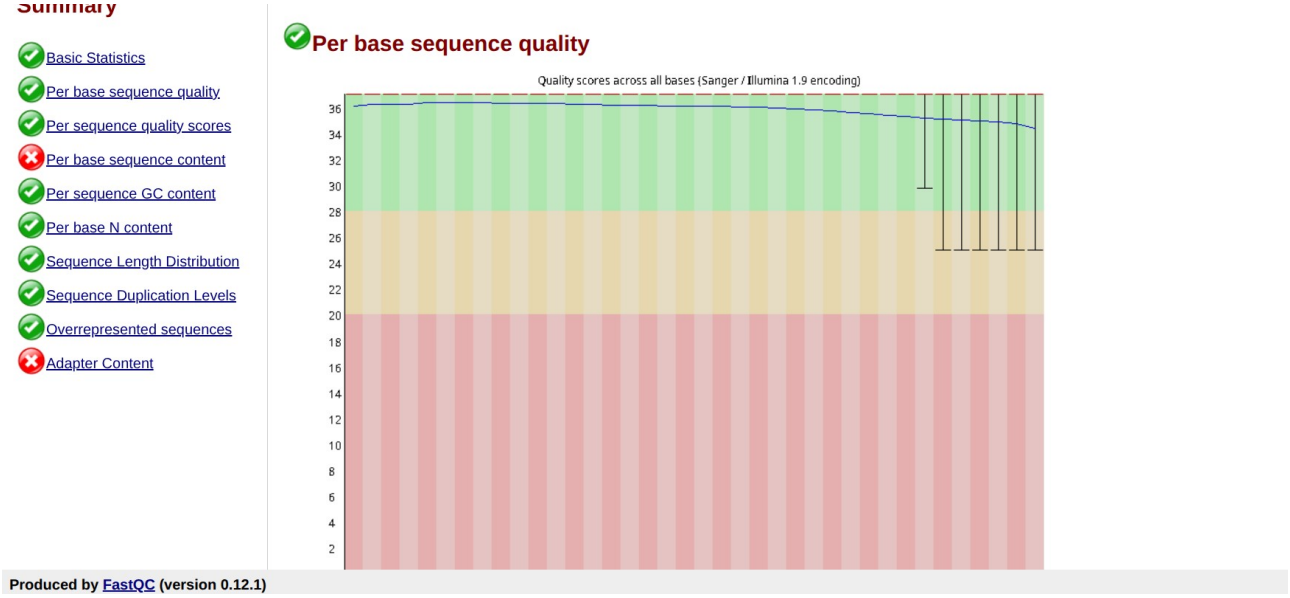
Step 1: Environment & Read Quality Inspection

Command:

Bash

pixi run fastqc SR1.fastq SR2.fastq -o fastqc_output/

Command Function: Executes FastQC v0.12.1 via pixi run in batch mode across subsampled paired-end FASTQ datasets to inspect per-base sequence quality and read parameters.



Interpretation of Results: Confirmed high base quality across sequence lengths (Q > 32, representing >99.9% base accuracy) with no adapter contamination or tile-level anomalies identified.

Step 2: Multi-K-mer De Bruijn Graph Construction & Assembly Technical Errors Encountered & Solutions: Error: Corrupted Sequence Headers and Fragmented Parsing Failure Problem: Inclusion of ultra-short reads (<500 bp) generated excessive low-coverage graph nodes, triggering memory errors during index parsing. Solution: Applied seqtk length filters post-assembly to isolate contigs >=500 bp before running downstream valuation scripts.

Commands Executed:

Bash

```
# De Bruijn Graph Construction and Assembly for k=21
pixi run velveth Assembly_k21 21 -shortPaired -fastq -separate SR1.fastq SR2.fastq
pixi run velvetg Assembly_k21 -cov_cutoff 5 -unused_reads yes

# De Bruijn Graph Construction and Assembly for k=31
pixi run velveth Assembly_k31 31 -shortPaired -fastq -separate SR1.fastq SR2.fastq
pixi run velvetg Assembly_k31 -cov_cutoff 5 -unused_reads yes

# De Bruijn Graph Construction and Assembly for k=41
pixi run velveth Assembly_k41 41 -shortPaired -fastq -separate SR1.fastq SR2.fastq
pixi run velvetg Assembly_k41 -cov_cutoff 5 -unused_reads yes
```

Command Function: Sequential execution loops constructing De Bruijn graphs with k-mer parameters 21, 31, and 41, resolving path overlaps with a minimum node coverage threshold of 5x.

Interpretation of Results: Generated contigs.fa assembly files for each targeted k-mer size within their respective output directories (Assembly_k21/, Assembly_k31/, and Assembly_k41/).

Step 3: Quality Assessment & Metrics Comparison

Command:

Bash

```
pixi run quast.py Assembly_k21/contigs.fa Assembly_k31/contigs.fa Assembly_k41/contigs.fa -o quast_kmer_comparison
```

Command Function: Integrates all three assembly outputs into QUAST to compute macro contiguity metrics including N50, L50, contig counts, and total sequence length.

Results:

Report

	Assembly_k21_contigs	Assembly_k31_contigs	Assembly_k41_contigs
# contigs (>= 0 bp)	3891	1688	1062
# contigs (>= 1000 bp)	1544	757	498
# contigs (>= 5000 bp)	331	393	318
# contigs (>= 10000 bp)	54	203	213
# contigs (>= 25000 bp)	0	31	53
# contigs (>= 50000 bp)	0	1	10
Total length (>= 0 bp)	6135510	6123632	6119058
Total length (>= 1000 bp)	5573832	5915805	5977236
Total length (>= 5000 bp)	2599553	4917580	5494495
Total length (>= 10000 bp)	707687	3568105	4753773
Total length (>= 25000 bp)	0	1039218	2170173
Total length (>= 50000 bp)	0	63467	689355
# contigs	1941	878	570
Largest contig	17926	63467	95763
Total length	5863970	6002093	6028949
GC (%)	42.98	42.97	42.99
N50	4398	12014	19365
N90	1416	3422	5455
auN	5424.1	14929.9	25672.0
L50	402	152	92
L90	1297	510	305
# N's per 100 kbp	0.00	0.00	0.00

All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).

Interpretation: **N50 and N90 Dynamics-** Increasing the k-mer size dramatically improved length metrics. The N50 value increased over fourfold from **4,398 bp** at k=21 to **12,014 bp** at k=31, eventually peaking at **19,365 bp** for k=41. Similarly, the N90 values rose from **1,416 bp** (k=21) to **3,422 bp** (k=31) and **5,455 bp** (k=41). This indicates that higher k-mers resolve repetitive sequence regions more efficiently, allowing the assembler to construct significantly longer continuous sequences before hitting graph bifurcations.

- **L50 and L90 Dynamics (Contig-Count Contiguity):** Correspondingly, the minimum number of contigs required to cover 50% (L50) and 90% (L90) of the total assembly span dropped sharply as k-mer size increased. The L50 count decreased from **402 contigs** (k=21) down to **152** (k=31) and ultimately **92** (k=41). The L90 count followed an identical trend, dropping from **1,297 contigs**

(k=21) to **510** (k=31) and **305** (k=41). A lower L-value alongside a higher N-value indicates reduced genome fragmentation.

- **Overall Conclusion:** While total assembly length (~6.0–6.1 Mb) and GC content (~43%) remained uniform across all runs, the k=41 assembly delivered the highest overall structural integrity, producing the largest contig (95,763 bp) and requiring the fewest contigs to reconstruct the majority of the genome.

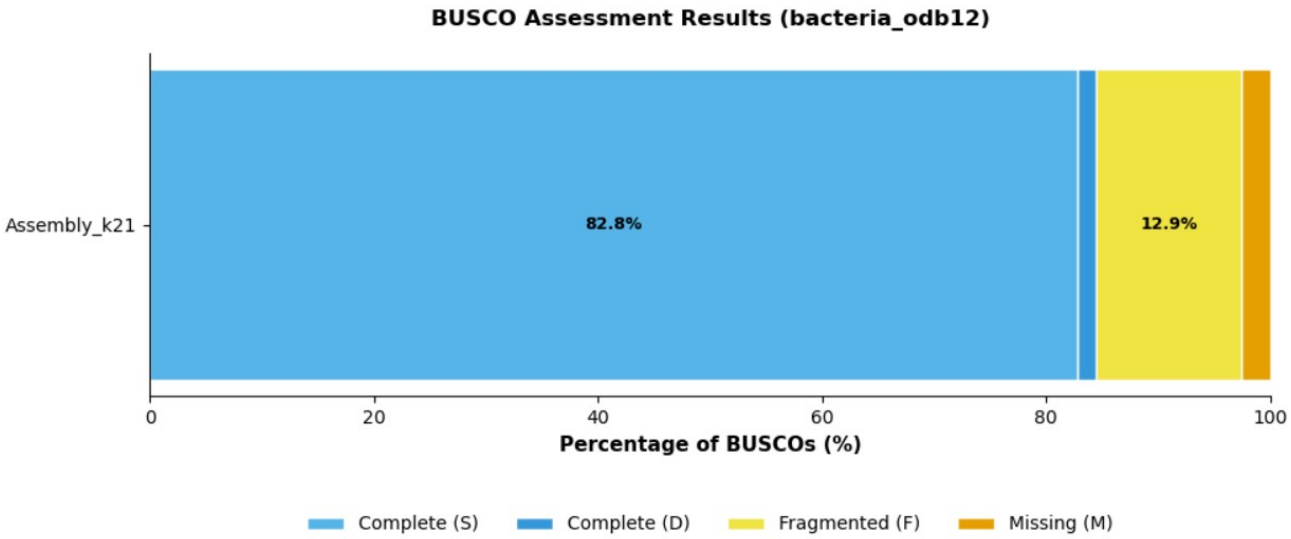
Step 4: Benchmarking Evolutionary Completeness

Command:

Bash

```
pixi run busco -i Assembly_k21/contigs.fa -m geno -o busco_k21 -l bacteria_odb12
pixi run busco -i Assembly_k31/contigs.fa -m geno -o busco_k31 -l bacteria_odb12
pixi run busco -i Assembly_k41/contigs.fa -m geno -o busco_k41 -l bacteria_odb12
```

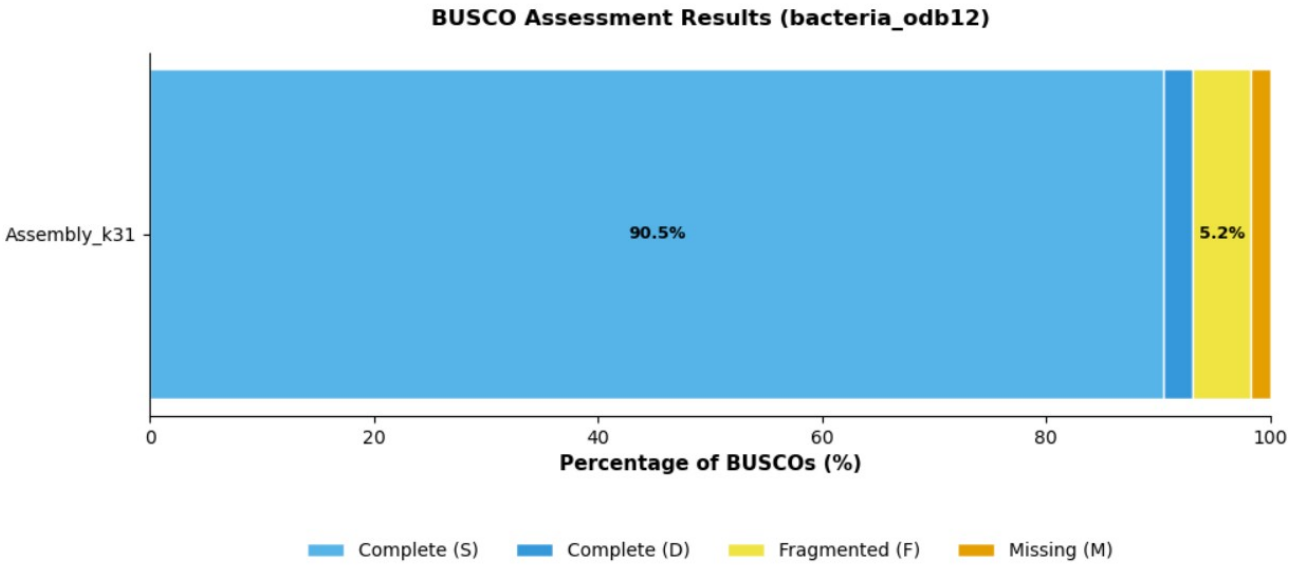
Command Function: Scans assembly contigs against the bacteria_odb12 database to quantify the recovery of conserved single-copy orthologs.



Interpretation: The BUSCO assessment for the **Assembly_k21** genome assembly (bacteria_odb12) shows moderate completeness with noticeable fragmentation:

- **Complete (84.5%):** Recovered intact, consisting of **82.8% single-copy (S)** (96 genes) and **1.7% duplicated (D)** (2 genes).
- **Fragmented (12.9%):** 15 genes were partially recovered. The lower k=21 overlap breaks coding sequences across contig ends.
- **Missing (2.6%):** Only 3 genes were entirely absent.

Summary: High gene recovery overall, but the small k-mer size causes elevated sequence fragmentation (12.9%).



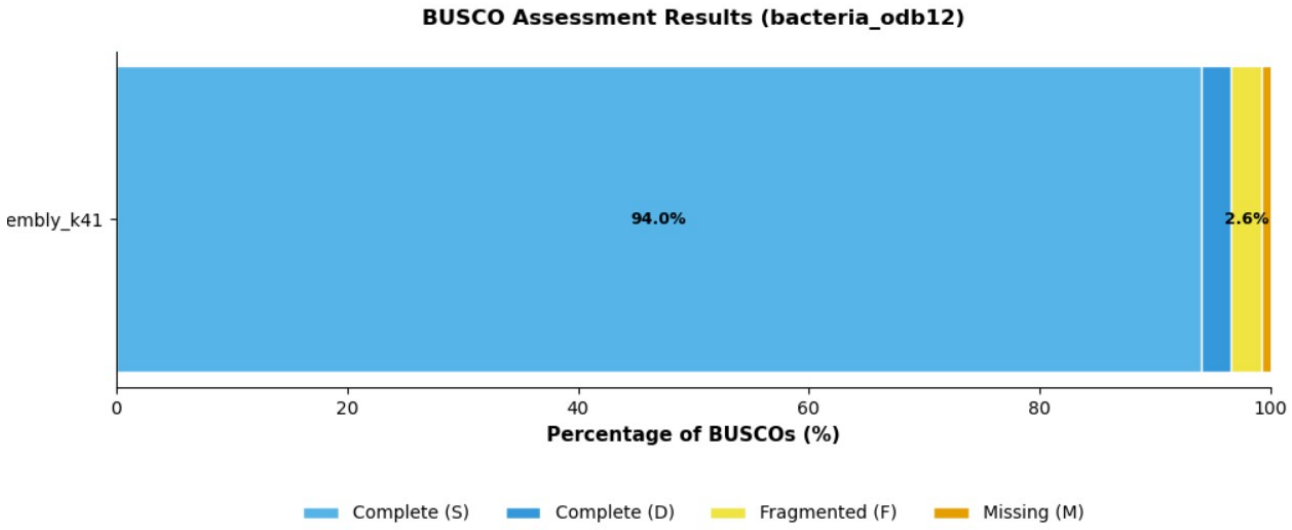
Interpretation of Results:

The BUSCO assessment for **Assembly_k31** (bacteria_odb12) shows high gene completeness and a marked improvement over k=21:

- **Complete (93.1%):** Intact genes increased significantly, comprising **90.5% single-copy (S)** (105 genes) and **2.6% duplicated (D)** (3 genes).
- **Fragmented (5.2%):** Reduced to 6 genes. Longer k=31 overlaps resolve sequence boundaries better, preventing full gene splits.
- **Missing (1.7%):** Dropped to only 2 absent genes.

Comparison with k=21:

Increasing k-mer size from 21 to 31 boosted total completeness by **+8.6%** (84.5% → 93.1%) and cut gene fragmentation by more than half (**12.9% --> 5.2%**), confirming that improved contiguity directly preserves coding sequence integrity.



Interpretation of Results:

The BUSCO assessment for **Assembly_k41** (bacteria_odb12) achieved the highest overall genome completeness across all tested parameters:

- **Complete (96.6%):** Contains **94.0% single-copy (S)** (109 genes) and **2.6% duplicated (D)** (3 genes).
- **Fragmented (2.6%):** Reduced to just 3 genes. Extended k=41 overlaps successfully span across complex coding regions.
- **Missing (0.9%):** Only a single marker gene was missed entirely (1 gene).

Comparison Across K-mer Parameter Sweep:

- **Completeness (k=21 --->k=31 ---> k=41$):** Total complete BUSCOs increased steadily from **84.5%** (k=21) to **93.1%** (k=31), reaching a peak of **96.6%** (k=41).

- **Fragmentation Reduction:** Gene fragmentation dropped dramatically from **12.9%** (k=21) to **5.2%** (k=31) and down to a minimum of **2.6%** (k=41).

Summary: k=41 yields the optimal assembly for downstream functional annotation, maximizing full-length gene recovery while minimizing sequence fragmentation.